

PMBOK / PMI | PMB-13

Schedule Baseline and Network Analysis

The approved schedule model, critical path and float analysis

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Schedule Baseline and Network Analysis
Document reference	PMB-13-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager
Approver	Sponsor
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-12	Activity and Milestone List	Input to the schedule
PMB-11	Schedule Management Plan	Governs this baseline
PMB-29	Earned Value Analysis Report	Measures against this baseline

How to use this template

Aspect	Guidance
Purpose	Records the approved version of the schedule model against which performance is measured, together with the network logic, critical path, float analysis and the assumptions on which the schedule depends.
Framework	PMBOK / PMI — Schedule Management — Develop Schedule
Created when	During planning, at the end of schedule development
Reviewed / updated	Through change control only; re-baselined only with approval
Owner	Project manager
Approver	Sponsor
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	The schedule model itself lives in the scheduling tool. This document records the baseline, the analysis and the reasoning — the parts the tool does not preserve.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- The critical path is identified and its drivers understood
- Schedule reserve is explicit and its ownership stated
- Resource levelling has been applied and its effect on the critical path recorded
- Assumptions and imposed dates are documented with their sources
- The baseline version and approval date are unambiguous

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1. Baseline details

Field	Detail
Baseline version	v1.0
Date baselined	18 Sep 2026
Approved by	L. Haddad, CISO
Scheduling tool and file location	Enterprise scheduling tool — Helios T2 master schedule, v1.0
Project start	01 Sep 2026
Project finish	30 Jul 2027
Total duration	4 weeks
Number of activities	12

2. Milestone baseline

Ref	Milestone	Baseline date	Type	Float available
MS-01	Programme mandate approved and funding released for Tranche 1	12 Mar 2026	Gate	Maintained by the P30 as part of the assurance evidence library
MS-02	Pilot cluster of 256 GPUs live in DXB-1 hall A	28 Aug 2026	Delivery	Enforced automatically in the delivery pipeline, with evidence retained
MS-03	Tranche 1 end-of-tranche review and Tranche 2 funding gate	18 Sep 2026	Gate	Dependent on the Enterprise Identity Programme (DEP-003)
MS-04	Hall B structural strengthening complete	15 Dec 2026	Delivery	Handled through the Agile Release Train rather than the predictive workstreams
MS-05	Utility uplift to 6 MVA energised	30 Nov 2026	External	Maintained by the P30 as part of the assurance evidence library
MS-06	Hall A scaled to 1,024 GPUs and accepted	26 Feb 2027	Delivery	Applies from hall A handover; hall B follows in Tranche 2 closure
MS-07	Platform v1.0 released — scheduling, quota, observability, image supply chain	31 Mar 2027	Delivery	Applies across DXB-1 halls A and B for the Tranche 2 scale build
MS-08	Independent security assurance and	23 Apr 2027	Assurance	Covers the 2,048-GPU

Ref	Milestone	Baseline date	Type	Float available
	penetration test passed			estate and the parallel filesystem tier

3. Critical path

The sequence of activities determining the earliest completion. Understand what drives it before deciding how to protect it.

Sequence	Activity ID	Activity	Duration	Driver
CP-01	Lodge utility uplift application	Execute floor strengthening works	3 working days	Constrained by the 6.2 MVA site limit at DXB-1 until 2028
CP-02	Install switchgear and energise 6 MVA supply	Lodge utility uplift application	2 hours	Aligned to the strategic theme of sovereign AI capability
CP-03	Install CDUs and secondary cooling loop	Install switchgear and energise 6 MVA supply	60 minutes	Scoped to the three onboarding tenants: AI Research, Fraud Analytics and Retail
CP-04	Build leaf-spine fabric in hall A	Install CDUs and secondary cooling loop	90 days	Escalated to the Portfolio Board where it exceeds programme tolerance

4. Near-critical paths

Paths with low float that will become critical if anything slips. These are where surprises come from.

Path	Total float	Activities	Risk
Confirmed at the Tranche 1 end-of-tranche review, 18 Sep 2026	Evidenced in the Tranche 2 gate pack, section 4	Scoped to the three onboarding tenants: AI Research, Fraud Analytics and Retail	Slow workload migration leaves the platform under-used (RSK-015)
Handled through the Agile Release Train	Constrained by the 6.2 MVA site limit at DXB-1 until 2028	Reflected in the solution intent as a fixed constraint	Fabric tuning falls short of the 92% all-reduce target (RSK-005)

Path	Total float	Activities	Risk
rather than the predictive workstreams			
Agreed with the integration partner at design freeze, 15 Oct 2026	Reflected in the solution intent as a fixed constraint	Constrained by the 6.2 MVA site limit at DXB-1 until 2028	Scarce platform engineering skills extend contractor dependence (RSK-004)
Applies only to workloads classified Restricted or above	Scoped to the three onboarding tenants: AI Research, Fraud Analytics and Retail	Evidenced in the Tranche 2 gate pack, section 4	Vendor allocation slippage defers the Tranche 2 benefit by a quarter (RSK-001)

5. Schedule compression applied

Technique	Where applied	Duration saved	Cost / risk incurred
Crashing	Applied to work package 3.1.	AED 640,000	AED 640,000
Fast tracking	Applied, with a design freeze gate added as a control.	No direct cost	No direct cost

6. Resource levelling

Constraint	Effect on schedule	Activities affected	Accepted by
Capital released tranche by tranche at gate approval only	Training throughput approximately 9% below the business case assumption	Agreed with the integration partner at design freeze, 15 Oct 2026	P. Raghavan, Platform Engineering Lead
Floor loading in hall B limited to 1,400 kg per rack position	Reportable data incident and loss of regulatory standing	Reflected in the solution intent as a fixed constraint	V. Chandra, Product Management
Grid capacity at DXB-1 is capped at 6.2 MVA until 2028	Critical path delay of up to ten weeks and a quarter of deferred benefit	Constrained by the 6.2 MVA site limit at DXB-1 until 2028	M. Verity, Head of P3O
Restricted training data must remain within UAE borders	Hall A capacity capped at 1,536 GPUs until resolved	Handled through the Agile Release Train rather than the predictive workstreams	G. Lindqvist, Service Operations Lead

7. Schedule reserve

Reserve type	Amount	Location in schedule	Owner	Release conditions
Contingency reserve	AED 9,400,000, derived from expected value of identified risks; drawn by the Programme Manager against a named risk	Tranche 2 completes 30 Jul 2027; tolerance of +4 weeks at programme level	Project manager	Performance bond of 10%, step-in rights, and design documentation held in escrow
Management reserve	AED 5,200,000, held outside the cost baseline and released by the SRO only	Tranche 2 completes 30 Jul 2027; tolerance of +4 weeks at programme level	Sponsor	Performance bond of 10%, step-in rights, and design documentation held in escrow

8. Imposed dates and their sources

Milestone	Imposed date	Source	Consequence of missing it	Negotiable?
Reviewed at each PI boundary by the Release Train Engineer	18 Sep 2026	Configuration management database	Critical path delay of up to ten weeks and a quarter of deferred benefit	Delegated to the Programme Manager within the agreed tolerance
Covers the 2,048-GPU estate and the parallel filesystem tier	25 Jun 2027	Assurance evidence pack, Tranche 2 gate	Hall A capacity capped at 1,536 GPUs until resolved	Enforced automatically in the delivery pipeline, with evidence retained
Maintained by the P3O as part of the assurance evidence library	28 May 2027	ALM tool — Helios ART board	Critical path delay of up to ten weeks and a quarter of deferred benefit	Reviewed monthly at the Programme Board

Milestone	Imposed date	Source	Consequence of missing it	Negotiable?
Subject to closure of ISS-003 before multi-tenant go-live	23 Apr 2027	Solution intent wiki, DXB-1 platform space	Reportable data incident and loss of regulatory standing	Excluded from Tranche 2 and deferred to the AUH-2 second site

9. Schedule assumptions

Ref	Assumption	Effect on schedule if wrong
ASM-001	Vendor GPU allocation of 2,048 units for Tranche 2 will be honoured at the quoted delivery dates	Tranche 2 completion slips by at least one quarter and contract terms are renegotiated
ASM-002	DXB-1 has 6 MVA of usable capacity available to the programme without a utility upgrade	Tranche 2 capped at 1,024 GPUs and the AUH-2 site is brought forward two quarters
ASM-003	Three business units will migrate committed workloads within six months of hall A handover	Utilisation benefit of AED 14.2m over five years is not realised in full
ASM-004	Training data classified as Restricted must remain within UAE borders at rest and in transit	Not applicable — this is a fixed constraint on design
ASM-005	Liquid cooling maintenance can be absorbed by the existing facilities contract without an uplift	Annual run cost increases by approximately AED 780,000
ASM-006	No production change may be made during the Group financial close window, the last five working days of each quarter	Not applicable — planned into the release calendar

10. Re-baseline history

Version	Date	Reason	Change request ref	Approved by
v1.0	18 Sep 2026	Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	Defer confidential computing enabler from Tranche 2 to Tranche 3	D. Okonkwo, Programme Manager
v1.0	15 Oct 2026	Supply chain provenance for AI tooling is weak; admission control is the only enforceable point	Integration partner claim for additional structural works	K. Adeyemi, Security Architect
v1.0	30 Nov 2026	Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is not yet justified	Bridge the hall B row 3 service void rather than relocate racks	I. Petrova, Data Platform Lead
v1.0	15 Dec	Physical lead times do not tolerate emerging	Dual-source 20%	J. Ferreira, Network

Version	Date	Reason	Change request ref	Approved by
	2026	<i>requirements, so this element is frozen before order</i>	<i>of Tranche 2 accelerator volume</i>	<i>Lead</i>